

STAT0023 ICA2 Report

Group 24

2025-04-25

Introduction

The 2016 United Kingdom referendum on European Union membership resulted in 51.2% of voters opting to leave, although voting patterns varied substantially across electoral wards. This analysis investigates the ward-level drivers of Leave support using data from 803 UK wards with complete referendum results, linked to demographic and socioeconomic covariates drawn primarily from the 2011 UK Census. The objective of this report is to explore how selected covariates are associated with Leave voting and to develop a statistical model to predict Leave support in the unobserved wards.

Exploratory analysis

Dataset Description and Preparation

The dataset includes 803 wards with known Leave results and 267 wards with missing values, retained with Leave set to NA for prediction purposes. Each row corresponds to one ward and includes education, age structure, ethnicity, employment, housing, and regional variables. Categorical variables (**RegionName**, **AreaType**) were converted to factors. Four variables—**ID**, **Leave**, **NVotes**, and **Postals**—are either identifiers or outcome-related and are excluded from the predictor set. **Postals** was removed based on BBC warnings about data accuracy concerns.

Analysis of the Response Variable

The response variable **LeaveProp** is defined as the proportion of Leave votes (**Leave** / **NVotes**). All values are confirmed to lie within the (0,1) interval, with the observed range approximately from 12.2% to 79.0%. Figure 1 displays the distribution of **LeaveProp**, which is moderately right-skewed, with most wards clustering around 0.6 and no extreme outliers observed. A Q-Q plot of **LeaveProp** (Figure 2) shows deviations from normality, particularly at both tails. This suggests that the residuals are unlikely to follow a normal distribution, and variance may not be constant, suggesting potential issues with normality assumptions in later statistical models.

Covariate Selection and Exclusion Criteria

The dataset contains 45 potential variables, including 43 numerical and 2 categorical covariates (**AreaType** and **RegionName**). Based on literature and correlation analyses, several covariates were selected for detailed exploration, while highly correlated variables (with absolute correlation $|r| \geq 0.9$) were excluded to reduce multicollinearity and improve interpretability.

For categorical covariates, **RegionName** identifies the specific region where the electoral ward is located, while **AreaType** classifies the type of administrative area for the ward. Due to high correlation between them, only **RegionName** was retained, as it better captures UK voting patterns¹.

RoutineOccupOrLTU and **HigherOccup** were excluded for their strong correlation with social grade or education variables. Aggregated class measures (**C1C2DE**, **C2DE**) provided in the original dataset were excluded in favour of individual social grade variables (**C1**, **C2**, **DE**), which we derived to enable clearer interpretation and assessment of the distinct effects associated with each social grade. Finally, **Deprived** also dropped in favour of the composite index **MultiDepriv**.

Demographic Covariate (Education, Age, Ethnicity)

Rosenbaum (2017)² emphasised that differences in education, age, and ethnicity explained much of the variation in Leave voting across UK wards. To explore the impact of education, we grouped the population into three levels: No Qualifications (**NoQuals**), Level 1 Qualifications (**L1Quals**), and Degree or Higher (**L4Quals_plus**). Figure 3 shows that **NoQuals** has a strong positive relationship with the Leave proportion. **L1Quals** shows a weaker positive trend, while **L4Quals_plus** is negatively associated with Leave votes. This suggests that higher education levels are linked to lower support for leaving the EU.

Age effects were analysed using four groups: children and adolescents, young voters, working-age Voters, and the elderly. **MeanAge** and **AdultMeanAge** were excluded due to redundancy. Figure 4 shows that working-age voters (aged 30–64) have the highest median Leave proportion compared to younger and older groups, suggesting they were more likely to support Leave.

For ethnicity, we focus on five categories: **White**, **Black**, **Indian**, **Pakistani**, and **Asian**. Figure 4 shows that Leave support varies significantly by ethnicity. Wards with predominantly **White** populations have the highest median support, well above 80%, while the other four groups all have medians below 10%.

Socioeconomic and Housing Covariates

To explore housing’s effect on voting behaviour, we examine how accommodation type relates to Leave voting (Figure 5). A higher proportion of homeowners is linked to more Leave support, while a higher share of renters is linked to less. **Owned** and **OwnedOutright** are strongly correlated and show similar trends, so only one will be used in the model. **SocialRent** and **PrivateRent**, which reflect different renter groups, may require further analysis before inclusion. Labour market vulnerability also influences Leave support through **Unemp**(unemployment). As shown in Figure 5, **Unemp** displays a mild upward trend, suggesting that wards with higher unemployment were more likely to support Leave. Population indicators (**Residents**, **Households**) and density (**Density**) were considered. Both **Residents** and **Households** serve as indicators of ward population size; we retained **Residents** for its more direct representation of the actual number of people. Additionally, **Density** (population per hectare) shows a generally negative correlation with **LeaveProp**. Figure 5 highlights that Leave support rises slightly at very low densities, suggesting stronger sentiment in rural areas, but declines steadily beyond 25%, indicating that denser areas tended to Remain, while sparser wards leaned Leave³.

Students and Multidimensional Deprivation

The variable **Students** displays a strong negative association between student concentration and **LeaveProp**, particularly within the 0–10% range (Figure 6). This supports claims that areas with higher student pop-

¹Denver, David, and Robert Johns. 2021. “The Geography of British Elections.” Springer EBooks, November, 201–27. https://doi.org/10.1007/978-3-030-86492-7_7.

²Rosenbaum, Martin. 2017. “Local Voting Figures Shed New Light on EU Referendum.” BBC News, February 6, 2017. <https://www.bbc.co.uk/news/uk-politics-38762034>.

³Riley, Alan, and Francis Ghilès. 2016. “Brexit: Causes and Consequences.” Cidob.org. 2016. <https://www.cidob.org/en/publications/brexit-causes-and-consequences>.

ulations tended to vote Remain⁴. Given the highly right-skewed distribution of this variable (as shown in the histogram), we applied a logarithmic transformation to better capture the relationship. Since **Students** is a percentage strictly greater than zero, the log transformation did not produce any missing values. The transformed plot reveals a smoother and more stable negative relationship across the full range. We also included **MultiDepriv**, a composite index capturing multiple forms of deprivation. As illustrated in Figure 5, it has a weak inverted-U relationship with **LeaveProp**: support increases with moderate deprivation, peaking around 30%, but slightly declines in the most deprived wards.

Social Grade Influence

We explored three social grade indicators: **C1** (junior white-collar workers), **C2** (skilled manual workers), and **DE** (unskilled and unemployed), which were explored individually to improve interpretability. As shown in Figure 5, **C1** exhibits a clear negative association with **LeaveProp**, suggesting that wards with a higher proportion of white-collar workers were more likely to vote Remain. In contrast, **C2** shows a strong positive relationship, indicating that skilled manual workers formed a significant support base for Leave. The pattern for **DE** appears non-linear, with Leave support rising at moderate levels but plateauing as the proportion increases further. These trends reflect the complex influence of social class on Brexit voting.

Model Building

GLM framework

Based on our EDA, the response variable shows a clear departure from normality, as evident from the Q-Q plot and histogram. This suggests that linear regression is unsuitable, and predictions could exceed the valid (0,1) interval. However, LOWESS curves of covariates against the response display approximate linearity, indicating a linear relationship may hold under a suitable link function. Consequently, we adopted a Generalised Linear Model (GLM) rather than a nonparametric Generalised Additive Model (GAM).

Choice of Distribution

Given that each vote represents an independent trial with probability p_i of being a ‘Leave’ vote, the number of ‘Leave’ votes can be assumed to follow a binomial distribution with parameters n_i (given by **NVotes** in each ward i) and p_i . Therefore, **LeaveProp** has a mean of p_i and a variance of $p_i(1 - p_i)/n_i$, satisfying binomial GLM assumptions. To account for varying ward sizes, we set **weights** = **NVotes**.

Choice of Link Function & Overdispersion

The logit link, being symmetric around $p_i = 0.5$, is appropriate as most **LeaveProp** values cluster near 0.5. It maps (0,1) intervals, ensuring that modelled probabilities remain within valid bounds. These properties make the log-odds scale effective at capturing the underlying relationships. While the logit was our initial choice, we also tested probit and complementary log-log links for comparison.

We first fit an initial model (**GLM_LeaveProp0**) with all EDA-suggested covariates. For **RegionName**, we apply sum-to-zero constraints, enabling the intercept to reflect the overall mean while region effects represent deviations. However, we observe a dispersion parameter of $45.2705 > 1$, indicating overdispersion, as the variance depends on p_i and is therefore not constant across wards. This may lead to unreliable standard errors and p -values. Additionally, the residual plot shows that many of the Pearson residuals exceed ± 2 , reinforcing the overdispersion.

⁴Graham-Harrison, Emma . 2016. “Young People on the EU Referendum: ‘It Is the End of One World, of the World as We Know It.’” The Guardian. June 26, 2016. <https://www.theguardian.com/politics/2016/jun/26/young-people-vote-anger>.

To resolve, we adopted a quasibinomial model introducing an additional dummy dispersion parameter. Among GLMs with logit, probit, and complementary log-log links (`GLM_Logit`, `GLM_Probit`, and `GLM_Cloglog`), the quasibinomial model with the logit link (`GLM_LeaveProp01`) had the lowest residual deviance (34970.7) and was selected for our subsequent model fitting.

Hierarchical Clustering

We simplified nine original `RegionName` levels into five new groups based on their estimated coefficients (Table 1). Under sum-to-zero contrasts, four group coefficients are estimated, with the fifth implied as the negative sum of the others. Although few group effects are insignificant (`GLM_LeaveProp02`), the new grouping captures distinct regional patterns.

Interaction

To capture regional variation, we introduce interactions between `NewGroup` and selected covariates. Since not all effects on the log-odds of voting ‘Leave’ differ by region, we start to include covariates supported by existing research to avoid overfitting. Carreras (2019)⁵ shows that population characteristics influence Leave support differently across UK regions, supporting the inclusion of a region-specific interaction for `Residents`. Spatial variation in the impact of ethnic diversity can justify ethnicity-related interactions⁶. The predictive power of education also varies substantially by region⁷, consistent with our findings for `NoQuals`, `L1Quals`, and `L4Quals_plus`. Similarly, the effect of deprivation is found to be stronger in some regions (e.g., the North East) than in others (e.g., London)⁸, supporting the inclusion of `MultiDepriv` interactions.

In addition to the contextual evidence, we plotted the `LeaveProp` against each covariate used in `GLM_LeaveProp02`, coloured by region and overlaid with region-specific GLM regression lines. This confirms regional slope differences for `Age18_29` and `log(Students)` (Figure 7). Therefore, these covariates were included in interaction terms.

We sequentially update the model from `GLM_LeaveProp3` to `GLM_LeaveProp14`, adding one covariate and its interaction with the regional group at a time. At each step, we compare the updated model with the simpler one using ANOVA F-tests to determine whether the interaction significantly improves model fit. For most covariates, including `Residents`, `Age18_29`, `Black`, `Indian`, `Asian`, `White`, `MultiDepriv`, `NoQuals`, `L1Quals`, and `L4Quals_plus`, the p -values are below 0.05, justifying their interactions.

However, interactions for `Pakistani` and `log(Students)` were excluded due to p -value > 0.05 and negligible reduction in deviance. After exclusion, we compare the reduced model (`GLM_LeaveProp16`) with the undropped model (`GLM_LeaveProp15`) using ANOVA F-test. The resulting p -value ($0.07828 > 0.05$) suggests that the simpler model has a comparable fit and is therefore preferred.

Final Model

To achieve parsimony and prevent overfitting, we further removed statistically insignificant covariates including `Age0_17`, `Age_65plus`, `OwnedOutright`, `PrivateRent` and `log(Students)`, with each removal validated by ANOVA F-tests yielding p -value > 0.05 . The resulting final model (`GLM_LeaveProp21`) therefore retains 21 main predictors: `Residents`, `NewGroup`, `Households`, `Age18_29`, `Age30_64`, `White`, `Black`, `Indian`,

⁵Carreras, Miguel. 2019. “‘What Do We Have to Lose?’: Local Economic Decline, Prospect Theory, and Support for Brexit.” *Electoral Studies* 62 (December): 102094. <https://doi.org/10.1016/j.electstud.2019.102094>.

⁶Beecham, Roger, Aidan Slingsby, and Chris Brunsdon. 2018. “Locally-Varying Explanations behind the United Kingdom’s Vote to Leave the European Union.” *Journal of Spatial Information Science*, no. 16 (June). <https://doi.org/10.5311/josis.2018.16.377>.

⁷Jump, Calvert, and Jo Michell. 2019. “Education and the Geography of Brexit.” SSRN. June 25, 2019. <https://ssrn.com/abstract=3411493>.

⁸Haußmann, Charlotte, and Tobias Rüttenauer. 2023. “Material Deprivation and the Brexit Referendum: A Spatial Multilevel Analysis of the Interplay between Individual and Regional Deprivation.” *European Sociological Review* 40 (3): 479–92. <https://doi.org/10.1093/esr/jcad057>.

Pakistani, Asian, Owned, SocialRent, NoQuals, L1Quals, L4Quals_plus, Unemp, C1, C2, DE, Density, and MultiDepriv. It also includes region-specific interactions between NewGroup and selected covariates (Residents, Age18_29, Black, White, Indian, Asian, NoQuals, L1Quals, L4Quals_plus, and MultiDepriv).

Compared to the null deviance of 389846, the final model reduced deviance to 27401, with the proportion of deviance explained given by:

$$D = 1 - \frac{D_{GLM_{LeaveProp21}}}{D_{null}} = 1 - \frac{27401}{389846} \approx 0.9297$$

Diagnostic Check

To check the assumptions of our final model, we examined residual diagnostic plots (Figure 8). Most Pearson residuals are scattered randomly around zero, suggesting that linearity and homoscedasticity hold reasonably well. The Scale-Location plot shows a nearly flat smoothed line, indicating a consistent spread of residuals across fitted values. The Q-Q plot of residuals aligns closely with the diagonal for most quantiles, although the upper tail shows a mild but systematic upward deviation. Lastly, the Cook’s Distance plot reveals that most points have negligible influence, with only three observations 303, 309, and 481, standing out as potentially influential.

Interpretation

Our final model supports most of the associations observed in the EDA. It suggests that a higher proportion of Leave votes is linked to a greater number of households, larger shares of White, Black, and Asian residents, lower education (L1Quals), higher unemployment, and more C2 occupations. For example, holding all other covariates constant, a 5% increase in the unemployment rate raises the log-odds of the proportion of Leave votes by 19.01% since $\exp(0.0348 \times 5) - 1 \approx 0.1901$.

In contrast, Leave support tends to be lower in wards with more permanent residents, younger and working-age populations, higher proportions of Indian and Pakistani residents, greater homeownership and social renting rates, higher education levels (L4Quals_plus), more C1 or DE occupations, higher population density, and greater deprivation.

Particularly, contrary to the BBC’s report, our model finds no significant association between older voters (Age_65plus) and Leave support, as indicated by a high p -value. This may suggest that other demographic factors among older voters, such as education or ethnicity, may be more influential. Instead, wards with more younger (18–29) and working-age (30–64) residents consistently show lower support for Leave.

Moreover, many interaction terms show significant effects. For example, the coefficient for NewGroup4:NoQuals (0.0237) corresponds to an odds ratio of $\exp(0.0237) \approx 1.024$, implying each one-percentage-point rise in adults without qualifications increases the odds of voting Leave by roughly 2.4% higher than the cross-group average effect. This highlights a stronger impact of educational disadvantage in these areas.

Limitation

One limitation of our final model is that, while most Pearson residuals support model assumptions, systematic deviations in the upper tail of the Q-Q plot indicate poor fits for a few observations. Additionally, Cook’s Distance reveals several influential points that may disproportionately impact coefficient estimates. Finally, several main effects and interactions have relatively large p -values, adding complexity without substantial explanatory power.

Prediction

Using the final model, we predicted Leave vote proportions for 267 wards with missing outcomes and provided standard deviations of our prediction error in the output file.

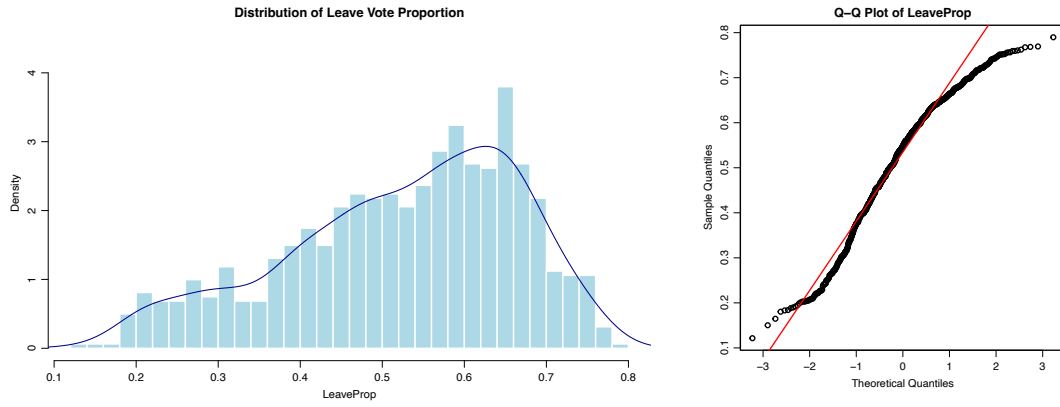


Figure 1: Histogram of LeaveProp & Figure 2: QQ Plot of LeaveProp

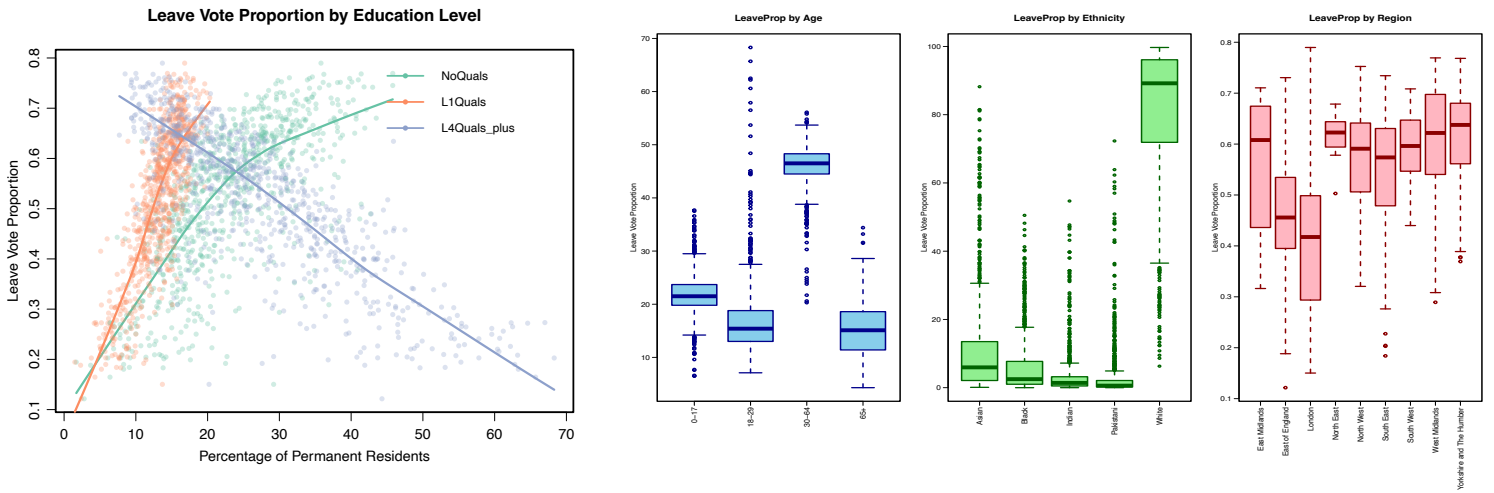


Figure 3: Scatterplot of NoQuals, L1Quals, and L4Quals_plus & Figure 4: Boxplots of age, ethnicity, and region

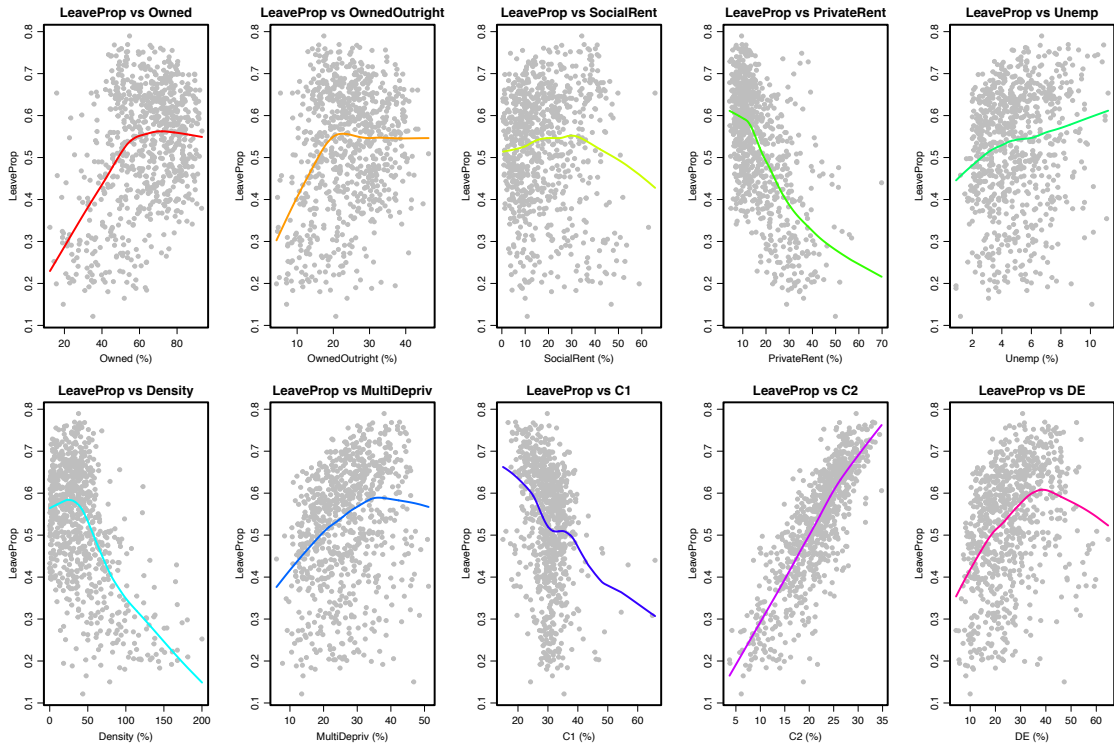


Figure 5: Scatterplots of key covariates

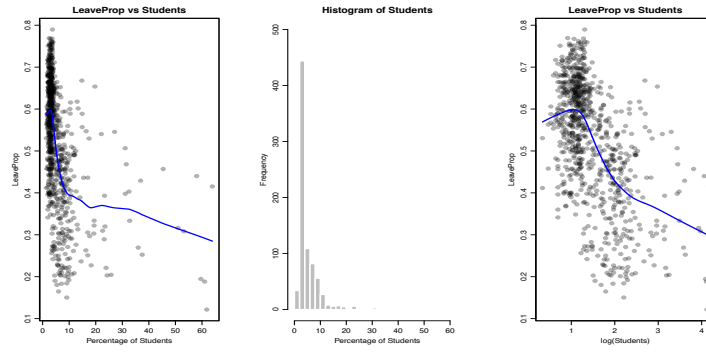


Figure 6: Scatterplots and histogram of students

Table 1: Regions in the five NewGroup clusters

Region	Cluster
NewGroup 1	
East Midlands	1
North East	1
South West	1
NewGroup 2	
East of England	2
NewGroup 3	
London	3
West Midlands	3
Yorkshire and The Humber	3
NewGroup 4	
North West	4
NewGroup 5	
South East	5

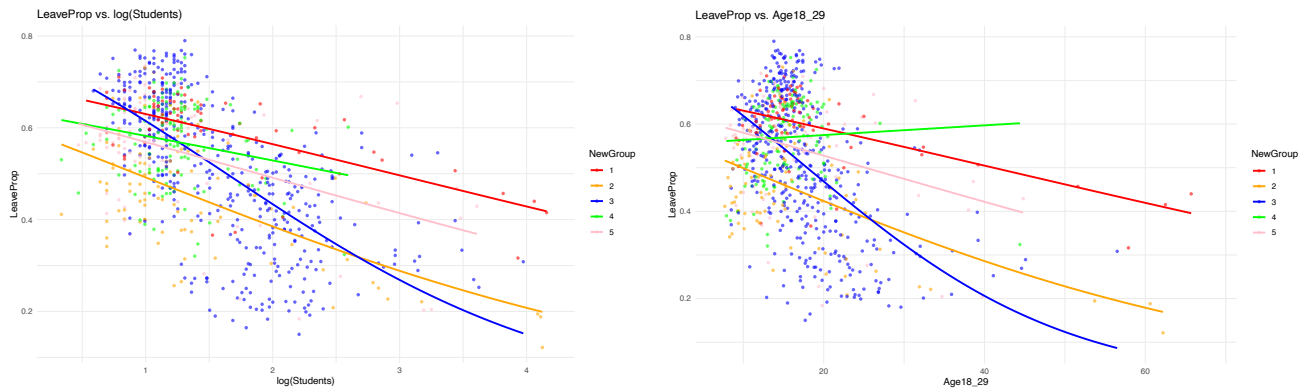


Figure 7: Interaction plots of $\log(\text{Students})$ and Age18_29

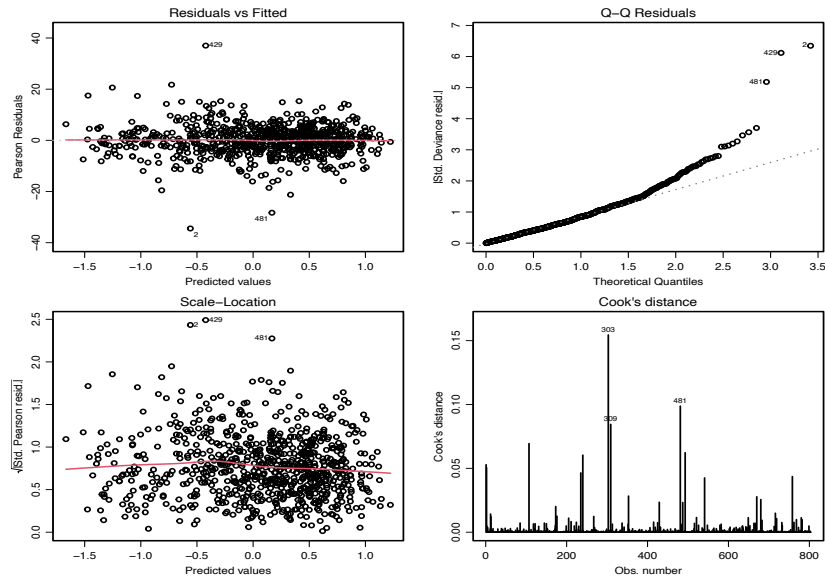


Figure 8: Diagnostic plots of our final model

Contribution

All group members (Student numbers: 22150953, 22167759, 22103901, 22031240) contributed equally to this ICA.